

PROJECT PITCH

ProjectMatch

AI-Powered Final Year Project Recommendation & Team Formation Platform

Problem

Every semester, faculty publish huge unstructured lists of project ideas across Google Docs and spreadsheets. Students manually scroll through hundreds of options, often duplicating ideas across teams, with no way to match their actual skills to a suitable project or find compatible teammates. There is no structured system — just a Google Form and a WhatsApp announcement.

Solution

A platform where faculty upload project ideas, students input their skills and interests, and an AI engine recommends the best-fit projects — while also suggesting compatible teammates based on complementary skills and flagging when too many teams are converging on the same idea.

Core Features

- Faculty Portal — upload and manage project ideas (title, description, domain tags, team size, capacity)
- Student Profile & Smart Matching — students list skills, stack, and interests; get ranked project recommendations via AI similarity matching
- Smart Team Formation — suggests teammates with complementary skills instead of random friend-groups
- Idea Deduplication Alerts — flags when multiple teams are picking near-identical ideas, encouraging diversity
- Admin Dashboard — department-wide view of all teams, ideas, domains, and approval status
- Status Tracking — replaces manual Google Form and spreadsheet tracking with a live approval workflow

Tech Stack

Layer	Technology
Frontend	React
Backend	Node.js + Express
Database	MongoDB
AI Layer	One LLM API (OpenAI / Gemini) for semantic matching — embedding-based similarity between student profile and project description; no model training required
Auth	JWT-based role auth (Student / Faculty / Admin)
Deployment	Render / Vercel + MongoDB Atlas

Team Split (4 Members)

Member	Responsibility
A	Faculty portal + idea CRUD + admin dashboard
B	Student profile system + AI matching / recommendation engine
C	Team formation logic + student-facing React UI

D	Auth (roles), notifications, deployment
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Why It's a Strong Pitch

- Real, self-experienced problem — directly inspired by this semester's own manual Google Form process
- Zero competition — no other team or faculty list has this idea
- Directly useful to DBUU — pitchable as a tool the department could actually adopt next semester
- Matches the team's stack exactly — MERN plus one lightweight AI layer, no exotic tech required